Lecture 1

Fundamentals of Numerical Analysis

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Course Info

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https://chhmou.github.io/math4610/
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- A. Greenbaum and T. P. Chartier, *Numerical Methods: Design, Analysis, and Computer Implementation of Algorithms*, Princeton University Press, 2012.
- R. L. Burden and J. D. Faires, *Numerical Analysis*, 10th Edition, Cengage Learning, 2015.
- L. N. Trefethen and D. Bau III, *Numerical Linear Algebra*, SIAM, 1997.

- Lectures: MWF 9:30–10:20am, Old Main 207
- Office: Geology 417B
- Office Hours: M: 1–2pm, R: 3–4pm, or by appointment
- email: changhong.mou AT usu.edu

Assessment

`https://chhmou.github.io/math4610/`

Grades:

Grades:	Homework (4 assignments, 12% each)	48%
	Quizzes	2%
	Midterm Exam	20%
	Final Project	30%

Homework:

- must acheive 50% to pass the course
- about weekly
- submitted by CANVAS or email
- in a report form
- source code included upon request
- no graphics by hand
- accepted for 50% credit up to one week late
- no dropped scores
- collaborate, but **do not** copy!
  - cite collaborators at the top
  - see departmental policy re: cheating

Finally. . .

https://chhmou.github.io/math4610/

Schedule and Notes:

- on the web
- everything is **tentative** including midterm exams
- "Good Questions worksheets" are lecture only

Questions?

Numerical Methods?

Numerical Analysis?

Numerical Calculation vs. Symbolic Calculation

- Numerical Calculation: involve numbers directly
  - manipulate numbers to produce a **numerical** result
- Symbolic Calculation: symbols represent numbers
  - manipulate symbols according to mathematical rules to produce a symbolic result

Example (numerical)

$$\frac{(17.36)^2 - 1}{17.36 + 1} = 16.36$$

Example (symbolic)

$$\frac{x^2 - 1}{x + 1} = x - 1$$

Analytic Solution vs. Numerical Solution

- Analytic Solution (a.k.a. symbolic): The exact numerical or symbolic representation of the solution
  - may use special characters such as π, e, or $\tan(83)$
- Numerical Solution: The computational representation of the solution
  - entirely numerical

Example (analytic)

$$\frac{1}{4}$$

$$\frac{1}{3}$$

$$\pi$$

$$\tan(83)$$

Example (numerical)

$$0.25$$

$$0.33333\ldots(?)$$

$$3.14159\ldots(?)$$

$$0.88472\ldots(?)$$

Numerical Computation and Approximation

- Numerical Approximation is needed to carry out the steps in the numerical calculation. The overall process is a numerical computation.

Example (symbolic computation, numerical solution)

$$\frac{1}{2} + \frac{1}{3} + \frac{1}{4} - 1 = \frac{1}{12} = 0.083333333\ldots$$

Example (numerical computation, numerical approximation)

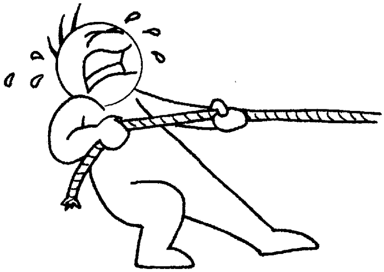
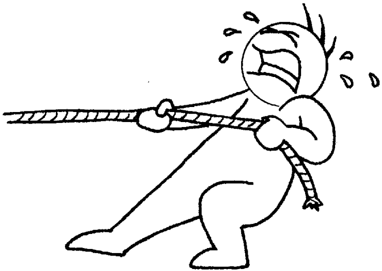
$$0.500 + 0.333 + 0.250 - 1.000 = 0.083$$

Method vs. Algorithm vs. Implementation

- Method: a general (mathematical) framework describing the solution process
- Algorithm: a detailed description of executing the method
- Implementation: a particular instantiation of the algorithm

- Is it a “good” method?
- Is it a robust algorithm?
- Is it a fast implementation?

The Big Theme

Accuracy

Cost

- date to 1650 BC: The Rhind Papyrus of ancient Egypt contains 85 problems; many use numerical algorithms (T. Chartier, Davidson)

- Approximates π with $(8/9)^2 * 4 \approx 3.1605$

History: Archimedes

- 287-212BC developed the ”Method of Exhaustion”
- Method for determining π
  - find the length of the permieter of a polygon inscribed inside a circle of radius $1/2$
  - find the permiter of a polygon circumscribed outside a circle of radius $1/2$
  - the value of π is between these two lengths

History: Method of Exhaustion

- A circle is not a polygon
- A circle **is** a polygon with an infinite number of sides
- C_n = circumference of an n-sided polygon inscribed in a circle of radius $1/2$
- $\lim_{n\to\infty} = \pi$
- Archimedes deterimined

$$\frac{223}{71} < \pi < \frac{22}{7}$$
$$3.1408 < \pi < 3.1429$$

- two places of accuracy....
- `http://www.pbs.org/wgbh/nova/archimedes/pi.html`

History: Method of Machin

- Around 1700, John Machin discovered the trig identity

$$\pi = 16 \arctan\left(\frac{1}{5}\right) - 4 \arctan\left(\frac{1}{239}\right)$$

- Led to calculation of the first 100 digits of π
- Uses the Taylor series of arctan in the algorithm

$$\arctan(x) = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} \ldots$$

- Used until 1973 to find the first Million digits

Numerical Analysis

Definition (Trefethen)

Study of algorithms for the problems of continuous mathematics

We've been doing this since Calculus (and before!)

- Riemann sum for calculating a definite integral
- Newton's Method
- Taylor's Series expansion + truncation

Big Questions

- How algorithms work and how they fail
- Why algorithms work and why they fail

- Connects mathematics and computer science
- Need mathematical theory, computer programming, and scientific inquiry

Numerical Analysis

A Numerical Analyst needs

- computational knowledge (e.g. programming skills)
- understanding of the application (physical intuition for validation)
- mathematical ability to construct and meaningful algorithm

Numerical Analysis

Numerical focus:

- **Approximation** An approximate solution is sought. How close is this to the desired solution?
- **Efficiency** How fast and cheap (memory) can we compute a solution?
- **Stability** Is the solution sensitive to small variations in the problem setup?
- **Error** What is the role of finite precision of our computers?

Why?

- Numerical methods improve scientific simulation
- Some disasters attributable to bad numerical computing (Douglas Arnold)
  - The Patriot Missile failure, in Dharan, Saudi Arabia, on February 25, 1991 which resulted in 28 deaths, is ultimately attributable to poor handling of rounding errors.
  - The explosion of the Ariane 5 rocket just after lift-off on its maiden voyage off French Guiana, on June 4, 1996, was ultimately the consequence of a simple overflow.
  - The sinking of the Sleipner A offshore platform in Gandsfjorden near Stavanger, Norway, on August 23, 1991, resulted in a loss of nearly one billion dollars. It was found to be the result of inaccurate finite element analysis.

I thought we were studying ”Numerical Methods”?!

Numerical analysis is the study of numerical methods

- Numerical Analysis: understanding general behavior of numerical methods
- Numerical Methods: understanding **how** to apply certain methods to solve specific tasks
- As programmers, we need to understand the concepts of numerical analysis and implementation aspects of the numerical method

We thus focus on

- Matlab implementation
  - fast learning curve
  - quick time-to-production: low development times
  - a major develop environment in scientific computing
  - integrated graphics
- Errors in computation
- specific methods for root finding, integrating, interpolation, etc.

Preliminaries...

- A computational experiment...
- relative vs absolute error
- Taylor
- nested multiplication
- Matlab

Goals

- To think like a numerical analyst
- Understanding error in numerical computing
- A look at the power of Tayor Series
- A brief look at the algorithmic importance in numerical methods
- Utilizing Matlab

Typical Process

Goal

Find $f'(1.0)$ for function $f(x) = e^x$.

Problem

Don't (want to need to) know $f'(x)$ explicitly.

Method

Use definition.

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Typical Process continued

Method

Use definition.

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

Listing 1: psuedo

```

h=1
x=1
for i=1 to 20 do
   h = h/2
   y = (f(x+h) - f(x))/h
   err = |(f(x) - y)|
end
```

Listing 2: Matlab

```
f=inline('exp(x)');
h=1;
x=1;
for i=1:20
  h=h/2;
  y=(f(x+h) - f(x))/h;
  err(i) = abs(f(x)-y);
end
```

Measuring Error

- Here we used the absolute error:

$$\begin{aligned}\text{Error}(x_a) &= |x_t - x_a| \\ &= |\text{true value} - \text{approximate value}|\end{aligned}$$

- This doesn't tell the whole story. For example, if the values are large, like billions, then an Error of 100 is small. If the values are smaller, say around 10, then an Error of 100 is large. We need the relative error:

$$\begin{aligned}\text{Rel}(x_a) &= \left|\frac{x_t - x_a}{x_t}\right| \\ &= \left|\frac{\text{true value} - \text{approximate value}}{\text{true value}}\right|\end{aligned}$$

- More on *types* of errors next time...

Taylor

- All we can ever do is add and multiply (more on the FPU later)
- We can't directly evaluate e^x, $cos(x)$, $\sqrt{x}$
- What to do? Taylor Series *approximation*

Taylor

The Taylor series expansion of $f(x)$ at the point $x = c$ is given by

$$f(x) = f(c) + (x-c)f'(c) + \frac{(x-c)^2}{2!}f''(c) + \cdots + \frac{(x-c)^n}{n!}f^{(n)}(c) + \ldots$$
$$= \sum_{k=0}^{\infty} \frac{(x-c)^k}{k!} f^{(k)}(c)$$